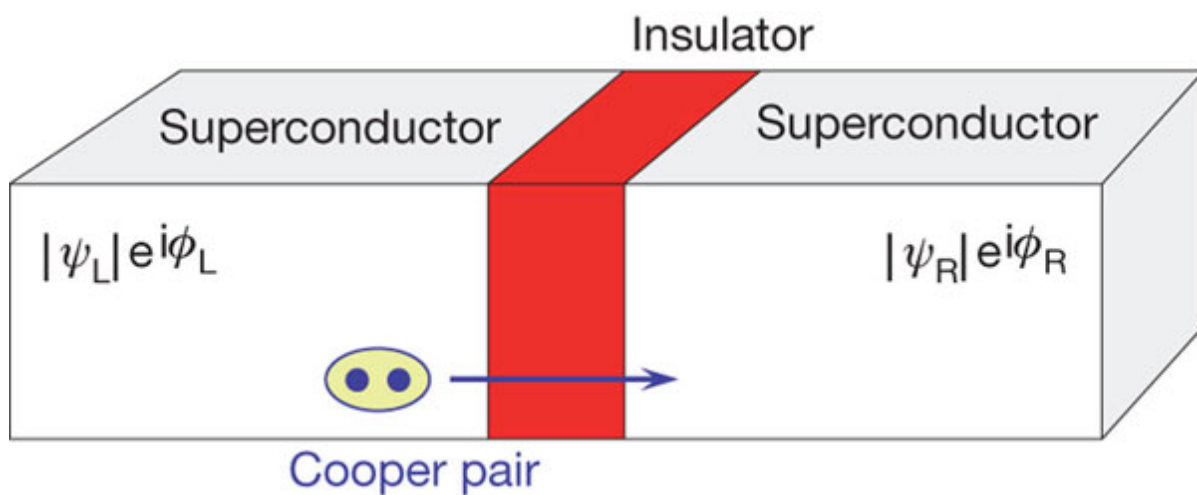




Project Outside the Course Scope

Transport properties of asymmetric Josephson junctions

Sagar Clemensen - fck916

Supervisors: Kasper Grove-Rasmussen and Asbjørn Drachmann

January 14, 2026

Abstract

In this project, the transport properties of asymmetric Al/AlO_x/NbN Josephson junctions were investigated and compared to those of symmetric Al/AlO_x/Al junctions. The measurements were carried out at a base temperature of $T = 150 \text{ mK}$ in a Kiutra cryostat using both current bias and voltage bias configurations. From measured IV characteristics, key transport parameters including the critical current I_c , the superconducting energy gap Δ/e , and the normal-state resistance R_n were extracted. For the symmetric junctions, the extracted energy gaps were consistent across devices, while the asymmetric junctions exhibited significantly larger variations between devices. For all devices the Ambegaokar-Baratoff relation deviated by one order of magnitude.

Additional voltage bias measurements were performed at elevated temperatures in order to study the temperature dependence of the superconducting energy gaps of aluminum and niobium nitride. From these measurements, Δ_{Al} and Δ_{NbN} were extracted and their temperature dependence analyzed. The critical temperature of aluminum was estimated and found to be in reasonable agreement with known values. Furthermore, subgap resistance measurements were used to estimate the Dynes parameter γ for selected devices.

Several junctions displayed transport behavior that diverged from theory, which is discussed in terms of possible device damage, or the presence of additional superconducting elements in series with the junctions. Overall, the results provide an experimental evaluation of the transport characteristics of asymmetric Josephson junctions and assess their consistency with theoretical expectations.

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1 Introduction

We are in the midst of the second quantum revolution, which aims to turn quantum mechanics into scalable and useful technologies. One of its central goals is the development of a fault-tolerant quantum computer. This objective is actively pursued at the Novo Nordisk Foundation Quantum Computing Programme (NQCP) at the Niels Bohr Institute (NBI). One of the leading platforms for developing such quantum computers is based on superconducting qubits. At the heart of these qubits are superconducting Josephson junctions.

These junctions are typically fabricated as symmetric Al/AlOx/Al structures, but asymmetric designs are also possible. In this project, asymmetric Josephson junctions consisting of Al/AlOx/NbN will be studied. These devices have been shipped from University of California, Berkeley.

The aim of this project is to study and evaluate the transport properties of these asymmetric Josephson junctions at $T = 150 \text{ mK}$. From the measured transport behavior, it should be possible to determine whether the junctions operate as intended. For comparison, symmetric Josephson junctions will also be measured, and additional measurements at higher temperatures will be performed.

2 Theory

2.1 Superconductivity

Electric resistivity ρ is a fundamental property of a material that measures how well it resists electrical current. In a metal, different scattering processes contribute to the resistivity. In a typical metal the three main scattering processes are impurities, electron-electron interactions, and electron-phonon collisions[1]. Scattering by impurities is generally independent of temperatures, whereas both electron-electron interaction and electron-phonon collisions are temperature dependent at low temperatures. At sufficiently low temperatures it is therefore expected that the resistivity of a metal takes the form

$$\rho = \rho_0 + aT^2 + \dots, \quad (2.1)$$

where $\rho_0 = \rho(T = 0)$ depends solely on the impurities in the metal[1].

While this model provides a good description of many metals, some materials exhibit fundamentally different behavior. When cooled below a specific temperature T_c , known as the critical temperature, their resistivity completely vanishes. These materials are called superconductors as electrical current can flow without dissipating. A superconductor is a distinct physical state and the critical temperature T_c is material dependent [1].

Although many superconductors are metals, other types of materials can also become superconducting. Certain elements such as oxygen exhibit superconductivity under sufficiently high pressure [1].

In this report, the superconductors aluminum (Al) and Niobium Nitride (NbN) are studied. The critical temperature of Al is 1.2 K [1], while NbN has a significantly higher critical temperature up towards 16 K [2].

Another defining property of superconductors is the Meissner effect. Superconductors are perfect diamagnets, so they expel magnetic fields as long as the field is lower than the

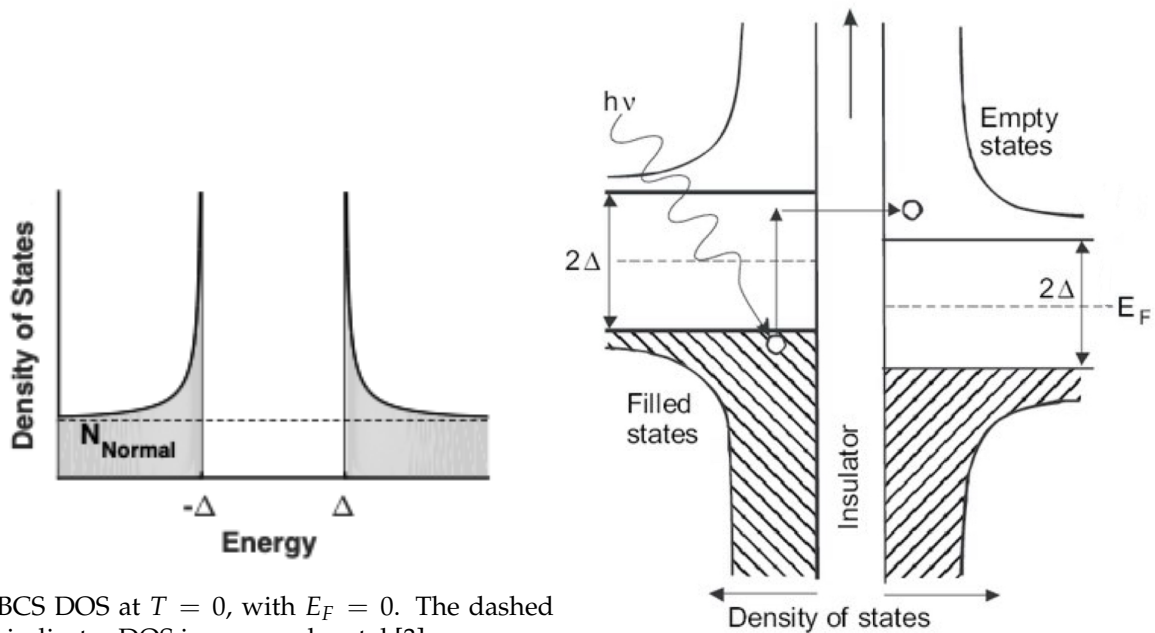
critical field, H_c . Just like T_c H_c is a material-dependent property [3].

Currents make self-fields so superconductors also have a critical current I_c . If a current larger than I_c passes through a superconductor it enters a finite-voltage state, similar to when the temperature rises above T_c or they are subject to a magnetic field stronger than H_c [3].

2.1.1 Cooper pairs

When a material is cooled below its critical temperature T_c , an effective attractive interaction caused by phonons becomes sufficiently strong to bind electrons into pairs. These electrons form Cooper pairs (CP) with opposite momentum and spin: $(\mathbf{k} \uparrow, -\mathbf{k} \downarrow)$. As a result, (CP) have spin zero and therefore behave as bosons. Being bosonic, they can condense into the same quantum ground state, which gives rise to dissipationless [3].

The pair potential or energy gap Δ is the energy scale associated with the formation of CP. It is on the order of $k_B T_c$, where k_B is the Boltzmann constant. The exact factor is predicted by Bardeen, Cooper, and Schrieffer (BCS) theory and depends on the specific superconducting material. Since the electronic states with the energetically most accessible electrons lie near the Fermi energy E_F , electrons within an energy window $E_F \pm \Delta$ have low enough energy to condensate into CP. As a consequence, an energy gap of width 2Δ opens in the electron density of states (DOS) as seen in fig. 2.1a. If a CP acquires an energy exceeding Δ the pair is broken and the electrons return to their normal state [3].



(a) BCS DOS at $T = 0$, with $E_F = 0$. The dashed line indicates DOS in a normal metal [3].

(b) DOS for a SIS-JJ. In order for a QP to tunnel it needs to acquire the energy $\Delta_1 + \Delta_2$. This picture shows a symmetrical JJ so $\Delta_1 = \Delta_2$, but that is not the case for an asymmetrical JJ. The picture has been modified from the original version [4].

Figure 2.1: DOS vs energy for a superconductor (a) and energy vs DOS for a JJ (b).

2.2 Josephson Junctions

A Josephson junction (JJ) consists of two superconductors separated by a thin non-superconducting barrier, which can be an insulator (SIS-JJ) or a normal metal (SNS-JJ). CP can tunnel between the two superconductors, giving rise to a supercurrent across the junction. The macroscopic

quantum state of each superconductor is described by a wavefunction, and the current across the junction is dependent on the phase difference. The current flowing in the junction is given by

$$I = I_c \sin(\theta_1 - \theta_2), \quad (2.2)$$

where θ_1 and θ_2 are the phases of the macroscopic wave functions on either side of the junction [1].

The critical current I_c is the maximum supercurrent that can flow across the junction. For currents $I < I_c$ the current across the JJ is perfectly dissipationless, but if current is driven to a higher value $I > I_c$, a finite voltage drop V develops across the junction. At these currents the JJ enters a finite-voltage state and the IV curve of the junction resembles that of a metal [1].

When in the finite-voltage state current is carried by quasi-particles since the CP have been broken. Fig. 2.1b indicates that QP need to acquire the sum of the two energy gaps $\Delta_1 + \Delta_2$ to tunnel through the barrier of a JJ. Therefore, in an IV curve such as that shown in fig. 4.3, the observed voltage gap corresponds to $\Delta_1 + \Delta_2$. Because the measurement is of voltage rather than energy, the gap appears at a value of $(\Delta_1 + \Delta_2)/e$. For asymmetrical JJs $\Delta_1 \neq \Delta_2$.

3 Method

3.1 Biasing

When sending an electronic signal through a component such as a semiconductor biasing is used to determine the properties of the signal. You can use either voltage biasing or current biasing. The signal also has to be sourced, and commonly voltage is sourced.

When using current biasing a load resistor is connected to the circuit. The load resistor has a resistance that is much greater than the resistance of the component you want to measure as well as the wires. Due to Ohm's law $V = IR$ this load resistance defines the current running through the system since the resistance of the component and wires are negligible compared to it. By using a big load resistor ie. $M\Omega$ you can source signals at a magnitude of V and still send signals at a order of μA or smaller. This greatly increases the signal to noise ratio.

When using voltage biasing you still source voltage, but this time a voltage divider is connected to the circuit. A voltage divider consists of two resistances R_1 and R_2 with R_2 being connected to ground. Given these two resistances and the sourced voltage V_s , the outgoing voltage $V_{out} = V_s R_2 / (R_1 + R_2)$. Often the physical component is listed with a division factor. This is given by $(R_1 + R_2) / R_2$. Using a voltage divider also greatly increases the signal to noise ratio. This time around the resistance in the system is defined by the resistance of the component plus wires and the current is once again defined by Ohm's law.

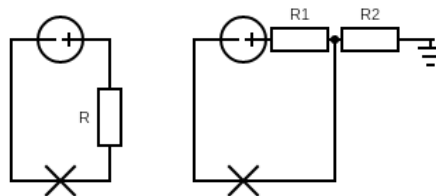


Figure 3.1: Circuit diagrams of current bias (left) and voltage bias (right). Typically a voltage divider is a single physical component. The x in the bottom of the diagrams is a JJ, but this setup works for all electrical components.

3.2 Electrostatic Discharge

Electrostatic Discharge (ESD) is a sudden flow of current that can happen between two objects with different electric potentials. One of the causes of ESD is through static electricity [5]. JJs are delicate objects and ESD is therefore at risk of destroying them. To mitigate this risk, a grounding cable can be used. The cable is connected via a breakout box to the same ground potential as the chip being measured. At the other end, a wrist strap is attached, when worn, it places the user at the same ground potential as the device, thereby reducing the risk of ESD damage. The grounding cable should be used whenever in contact with something that is connected to the device or the device itself.

3.3 Bonding

The JJs measured in this project are fabricated onto a chip with a 4-wire design for each JJ/device. In order to perform any measurements, the chip is placed onto a daughterboard and bonded to it. Aluminum wires are threaded to the edge of the daughterboard and then to one of the four contacts connected to every JJ. It is now possible to send and receive signals between the chip and the daughterboard. The daughterboard is connected to a motherboard on a puck, which is then loaded into the cryostat resulting in the chip being connected to the breakout box.

3.4 Setting up the experiment

The experiment was performed on a Kiutra cryostat. The cADR mode was used as measurements were done across several days. Measurements were performed on both symmetrical Al/AlOx/Al and asymmetrical Al/AlOx/NbN JJs fabricated with standard oxidation. In order to measure the signal across the JJ two Keysight DMM's are connected to the chip with BNC cables that are hooked up to a breakout box. Since the signal is small a voltage amplifier is used for measuring the voltage drop across the JJ. A Keithley sourcemeter applies a DC source voltage to the chip. Both current bias and voltage measurements are performed, so either a load resistor or voltage divider is connected between the source meter and breakout box. An I/V converter is connected to the drain contact of the breakout box amplifying the signal and allowing the DMM to measure the current across the JJ in units of Volts. Both the amplifier and I/V converter increase the signal to noise ratio.

For current bias a load resistor of either $100\text{ M}\Omega$ or $1\text{ M}\Omega$ was used. For voltage bias it was a voltage divider of either 1003.3 V/V or 10000 V/V .

Most of the measurements were performed at 150 mK . For both current- and voltage-bias configurations, measurements were carried out in the vicinity of the critical current for each JJ. The voltage was first sourced manually until a finite voltage drop developed across the JJ, thereby ensuring that the junction was in the finite-voltage state.

For current bias measurements, additional sweeps were performed with voltages reaching values up to $3 - 4\text{ mV}$, ensuring that the junction remained in the finite-voltage state over an extended bias range.

After manually determining the appropriate range of the sourced voltage, an automated sweep was performed using Python code on a computer connected to the Keithley source and the DMMs, which simultaneously recorded all measurement data. For measurements not focused on the temperature dependence of the JJs, both an up-sweep with increasing source voltage and a down-sweep with decreasing source voltage were conducted.

For device 3 on chip R4C2 measurements at several temperatures were performed. Starting at 150 *mK* and raising the temperature in increments of 150 *mK* up to 1.45 *K* voltage bias sweeps were conducted. A voltage sweep on device 9 from R4C2 was also performed at 1.3 *K*.

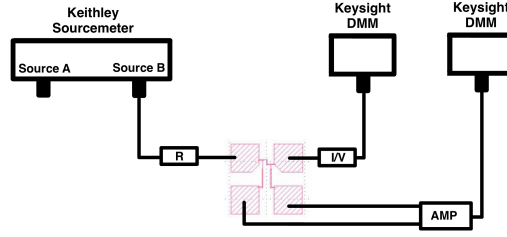


Figure 3.2: Schematic of the experimental setup for a current bias measurement. For a voltage bias measurement the load resistor R is changed to a voltage divider.

3.5 Chips measured

Three different chips were measured in the project labeled R2C4, R4C4, and R4C2. Each had 12 JJs that were fabricated as a modified Dolan bridge design where the lead 1 dimension was varied as either 100 *nm*, 150 *nm*, 200 *nm*, or 250 *nm*. The JJs were fabricated with three different gap sizes 250 μm , 300 μm , and 350 μm . All chip layouts are provided in appendix A.

4 Results

4.1 Measurements performed at low temperatures

Several electrical transport properties of the JJs were extracted from the measurements. By examining a current bias IV curve of the measured drain current and voltage drop across a JJ, its energy gap $(\Delta_1 + \Delta_2)/e$ and critical current I_c can be extracted. Even though the symmetrical JJs are Al/AlO_x/Al, the two Al leads are different thickness as shown in Appendix A fig. A.3. Therefore $\Delta_1 \neq \Delta_2$ for all devices. This was done for 8 different JJs which all had a single gap during the up-sweep and down-sweep. This is the expected behavior according to theory. Some of the measurements did not behave in this manner which is discussed in chapter 5. I_c was also extracted from the voltage bias measurements. Using the current bias measurements that reached voltages up to 3 – 4 *mV*, the normal-state resistance R_n can be extracted. All of these results are listed in table 4.1.

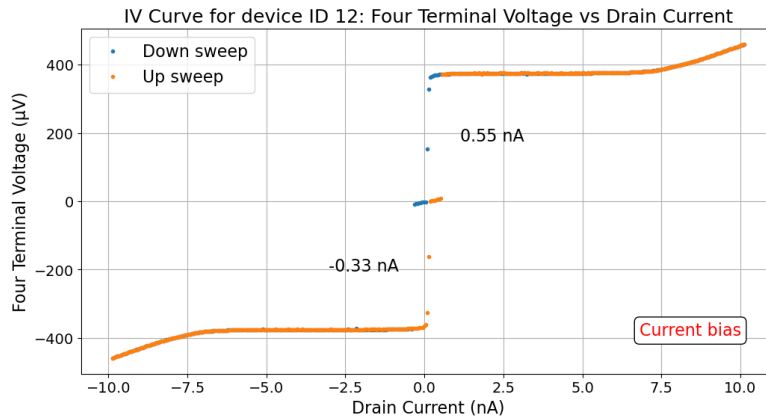


Figure 4.1: IV-curve of device 12 from chip R2C4 used for extracting I_c . The JJ is a symmetrical Al/AlO_x/Al junction.

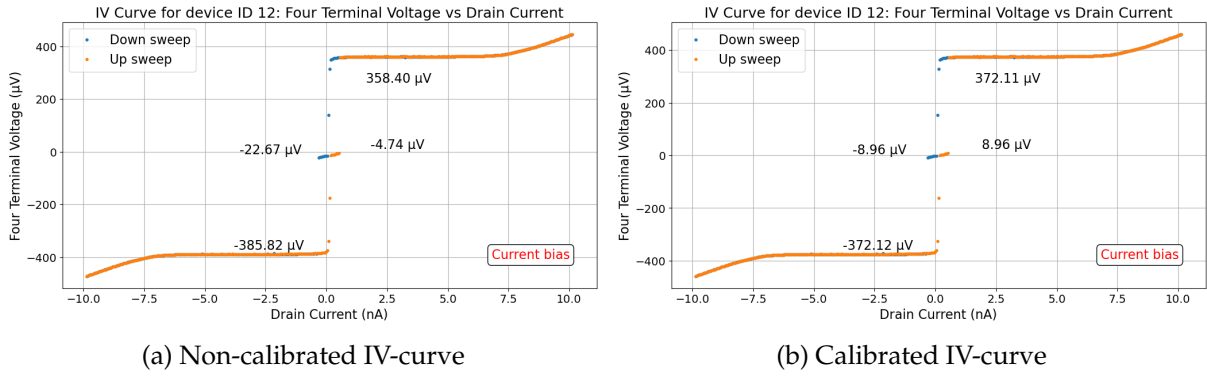


Figure 4.2: IV-curve of device 12 from chip R2C4 used for extracting the energy gap $(\Delta_1 + \Delta_2)/e$. The IV-curve shown in fig. 4.1 and all IV-curves used for extracting the data in chapter 5 been calibrated according to the manner shown between (a) and (b).

I_c is extracted as the x -value where the voltage jumps up or down as shown in fig. 4.1. For the extraction of $(\Delta_1 + \Delta_2)/e$ the data is first calibrated. The calibration is done by ensuring that the outermost values of the voltage before the gap for the up-sweep and down-sweep are equal. In the case of fig. 4.2a it is the values $-22.67 \mu\text{V}$ and $-4.74 \mu\text{V}$. The data is then aligned as shown in fig. 4.2b. This process is used for all IV-curves used for extracting the values shown in table 4.1.

After the calibration $(\Delta_1 + \Delta_2)/e$ is extracted as the y -value just after the jump. For fig. 4.2b the value is $372.1 \mu\text{V}$ for the up-sweep. In fig. 4.3 an IV curve for one of the asymmetrical JJs is shown. The procedure for calibrating the data and extracting I_c and $(\Delta_1 + \Delta_2)/e$ is exactly the same as for the symmetrical ones.

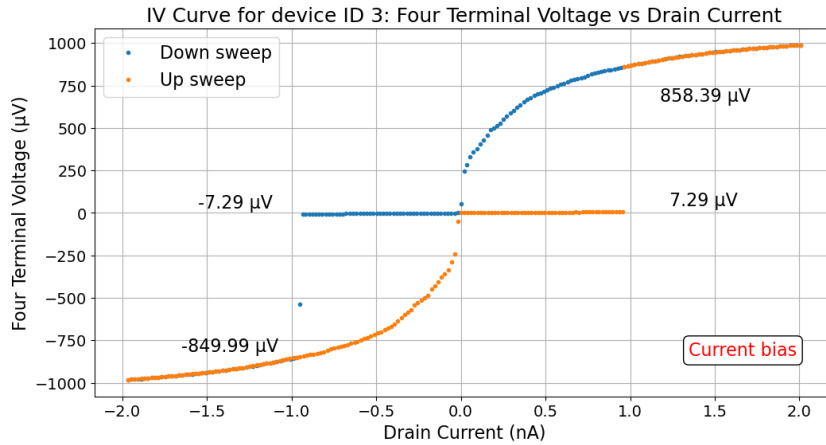


Figure 4.3: IV-curve of device 3 from chip R4C2 used for extracting $(\Delta_1 + \Delta_2)/e$. The JJ is an asymmetrical Al/AIOx/NbN junction.

It is possible to extract R_n from an IV-curve where the drain current is well above I_c . Here the JJ is in a finite-voltage state and by fitting Ohms law R_n can be extracted as the slope. R_n is extracted from both negative and positive current.

For the voltage bias IV curves I_c was extracted as the highest y -value of the peaks around $V = 0$. Since the voltage bias IV curves weren't used to extract $(\Delta_1 + \Delta_2)/e$ they weren't calibrated. In fig. 4.5 a full IV curve of a symmetrical JJ as well as a zoom in near $V = 0$ is shown together with the extracted I_c values.

For the asymmetrical JJs there were two peaks near $V = 0$. Here I_c was taken as the highest y -value of the first peak. Since the data had low resolution the data points were connected in order to see the peak. One such plot is shown in fig. 4.6.

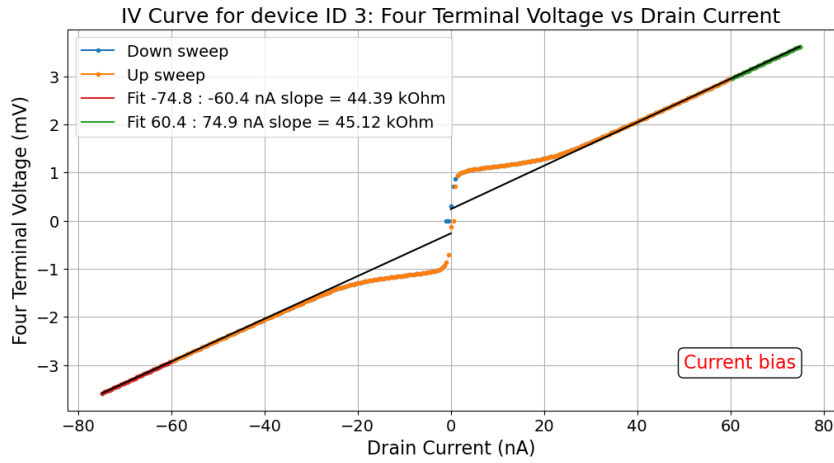


Figure 4.4: IV-curve of device 3 from chip R4C2 used for extracting R_n .

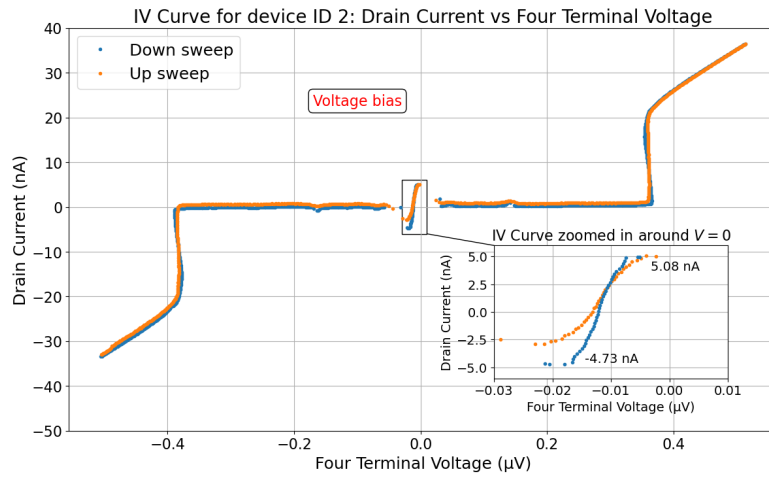


Figure 4.5: IV-curve of device 2 from chip R2C4 used for extracting I_c . The JJ is symmetrical.

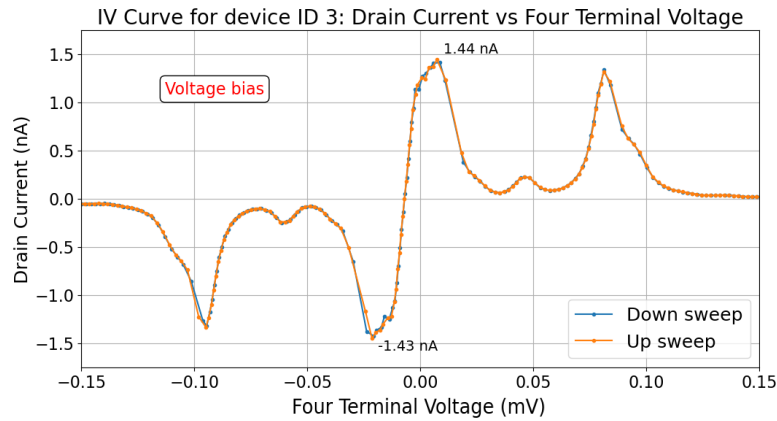


Figure 4.6: IV-curve of device 3 from chip R4C2 used for extracting I_c . The JJ is asymmetrical.

In order to get an error on the measured voltage a current bias IV curve near $V = 0$ was studied. A linear function was fitted to the measurements with $I < I_c$ and the residuals calculated. An error values was then determined by taking the standard deviation of the residuals, which was $0.1 \mu V$, as the error on voltage measurements. This error is used for the fit in fig. 4.4 and also used to determine the amount of significant digits for $(\Delta_1 + \Delta_2)/e$. In appendix B fig. B.1 a plot showing the fit and residuals can be seen.

Table 4.1: Summary of extracted parameters from the current bias and voltage bias measurements performed at 150 *mK*. The uncertainties on R_n arise during fitting. The values of I_c separated by the vertical line are from the voltage bias measurements while all the other values are from current bias. The horizontal line distinguishes between the different chips measured. The first row of measurements are from R4C4, the second section from R2C4, and the last from R4C2. In appendix B the remaining plots used to extract the parameters can be viewed.

ID	I_c [nA] Up	I_c [nA] Down	$\frac{\Delta_1+\Delta_2}{e}$ [μV] Up	$\frac{\Delta_1+\Delta_2}{e}$ [μV] Down	R_n [k Ω] Left (red)	R_n [k Ω] Right (green)	I_c [nA] Up	I_c [nA] Down
6	0.29	-0.30	832.1	-837.1	63.420 ± 0.002	65.551 ± 0.002	0.11	-0.12
1	1.78	-1.49	372.5	-373.6	21.238 ± 0.009	21.261 ± 0.009	2.17	-1.96
2	3.77	-3.62	372.7	-372.4	12.154 ± 0.003	12.142 ± 0.003	5.08	-4.73
4	7.58	-7.12	370.8	-371.0	7.373 ± 0.003	7.385 ± 0.003	8.59	-8.42
12	0.55	-0.33	372.1	-372.1	36.58 ± 0.02	36.61 ± 0.02	0.79	-0.57
3	0.96	-0.95	858.4	-850.0	44.390 ± 0.004	45.116 ± 0.004	1.44	-1.43
4	1.39	-1.39	880.1	-881.0	36.483 ± 0.004	36.600 ± 0.005	1.95	-2.01
9	1.61	-1.67	923.7	-927.0	32.638 ± 0.004	32.931 ± 0.004	2.40	-2.40

The extracted gaps $(\Delta_1 + \Delta_2)/e$ for the symmetrical JJs are all within 2 μV of each other as would be expected. This is not the case for the asymmetrical JJs. For each device the up and down sweep values are within 10 μV of each other, but for device 6 from R4C4 and device 9 from R4C2 the difference is roughly 100 μV .

For the extracted I_c up and down sweep value are close as would be expected. The I_c extracted from the voltage bias measurement varies a good amount from the current bias values. In all cases except for device 6 from R4C4 the voltage bias value is greater. Even though the values are different the relative order of greatest to lowest value between each device is the same for both current bias and voltage bias values.

According to Ambegaokar-Baratoff's theory $I_c R_n \approx 2\Delta/e$ for ideal symmetric SIS JJs [6]. This relation was applied to the extracted parameters and shown in table 4.2. For all devices, this relation deviates by one order of magnitude. Since this relation is for ideal symmetric SIS JJs, one order of magnitude is reasonable given the non-ideal junctions.

Table 4.2: Comparing Ambegaokar-Baratoff's theory for ideal symmetric SIS JJs to the values for $(\Delta_1 + \Delta_2)/e$. Here the average of the numerical values are used for $(\Delta_1 + \Delta_2)/e$ and I_c and the weighted average for R_n . The first column is devices from R4C4, the second from R2C4, and the third from R4C2.

Device ID:	6	1	2	4	12	3	4	9
$I_c R_n$ [μV]	19.0	34.7	44.9	54.2	16.1	42.7	50.8	53.8
$(\Delta_1 + \Delta_2)/e$ [μV]	834.6	373.1	372.5	370.9	372.1	854.2	880.8	923.4

In table 4.3 the weighted averages of R_n are compared to the room temperature resistance R_{room} measured by the fabrication team. The width of lead 1 and the gap size is also listed. A JJ with a smaller gap and a larger width is expected to have a lower resistance. R_{room} adheres to this principle, but not all values of R_n do. For example, based on the extracted values from R4C2, the R_n of device 9 is lower than that of device 4, despite device 9 having a larger gap.

Looking at the value of R_n from device 6 chip R4C4 and comparing it to those from R4C2, it seems that having a larger width is more important than a smaller gap. Device 6 has the

lowest lead width and gap size and the highest R_n and R_{room} .

The measurements of R_n from R2C4 are all consistent with a larger width and smaller gap resulting in a lower resistance.

For all measurements $R_n > R_{room}$. For the asymmetrical JJs R_n is several times greater than R_{room} .

Table 4.3: Weighted average of extracted R_n compared to room temperature measurement, lead 1 width and gap size. The horizontal line distinguishes between the different chips measured. The first row of measurements are from R4C4, the second section from R2C4, and the last from R4C2.

ID	R_n [k Ω]	R_{room} [k Ω]	Lead 1	Gap
6	64.486 ± 0.001	18.398	150 nm	250 μm
1	21.250 ± 0.006	13.038	100 nm	300 μm
2	12.148 ± 0.002	8.839	150 nm	300 μm
4	7.379 ± 0.002	5.022	250 nm	300 μm
12	36.593 ± 0.012	22.204	100 nm	350 μm
3	44.753 ± 0.003	11.396	200 nm	300 μm
4	36.541 ± 0.003	9957	250 nm	300 μm
9	32.784 ± 0.003	11.931	250 nm	350 μm

4.2 Measurements performed at higher temperatures

By performing voltage bias sweeps on device 3 from R4C2, an asymmetrical JJ, the temperature dependence of Aluminum's energy gap Δ_{Al} and Niobium Nitride's energy gap Δ_{NbN} can be extracted. This was done by numerically calculating the differential conductance dI/dV from the measured current, and voltage across the JJ and extracting the voltage at which peaks occur in the differential conductance. The extracted voltages correspond to the midpoints of the peaks, which were determined by eye. Consequently, these values carry a substantial uncertainty.

The peaks occurring at $V > 1$ mV were identified as corresponding to $(\Delta_{NbN} + \Delta_{Al})/e$ and those at 0.6 mV $< V < 0.9$ mV as corresponding to $(\Delta_{NbN} - \Delta_{Al})/e$ [7]. From $T = 750$ mK and below, the peaks at $(\Delta_{NbN} - \Delta_{Al})/e$ start spreading into two peaks, so they were not used. The peaks at $(\Delta_{NbN} + \Delta_{Al})/e$ were still extracted as was Δ_{Al} by assuming that Δ_{NbN} stays constant at low temperatures. The average of the previously obtained values of Δ_{NbN} were used as the constant value. The measurements are shown in fig. 4.7. Only up-sweeps were performed.

The extracted gaps and their temperature dependency are shown in fig. 4.8. Since NbN has a higher T_c than Al Δ_{NbN} stays constant while Δ_{Al} becomes smaller with increasing T .

The temperature dependency of Δ can be approximated by the following equation [3]¹

$$\Delta(T) = \Delta(0) \cdot \tanh \left(1.74 \sqrt{\frac{T_c}{T} - 1} \right). \quad (4.1)$$

In fig. 4.8b the measurements of Δ_{Al} are fitted with eq. (4.1). The values of Δ_{Al}/e come with a systematic uncertainty do to the method used for extracting them. To quantify this an increasing error was used for the fit until $p(\chi^2, N_{dof}) \geq 0.05$. This was the case for an error

¹This equation has been slightly modified from the cited thesis, since a mistake was found.

$\sigma_{\Delta_{Al}} = 3.4 \mu\text{eV}$. The first three measurements were not included in the fit as they resulted in a significantly higher χ^2 . The fit gives $\Delta(0) = (241 \pm 2) \mu\text{eV}$ and $T_c = (1.46 \pm 0.01) \text{ K}$. For Aluminum the official value of is $T_c = 1.2 \text{ K}$ and the value extracted in [7] is $\Delta_{Al} = 200 \mu\text{eV}$.

For Niobium Nitride the averaged value came out to be $\Delta_{NbN} = (899 \pm 3) \mu\text{eV}$, which is quite far from the value found in [7] of $\Delta_{NbN} \approx 1.4 \text{ meV}$.

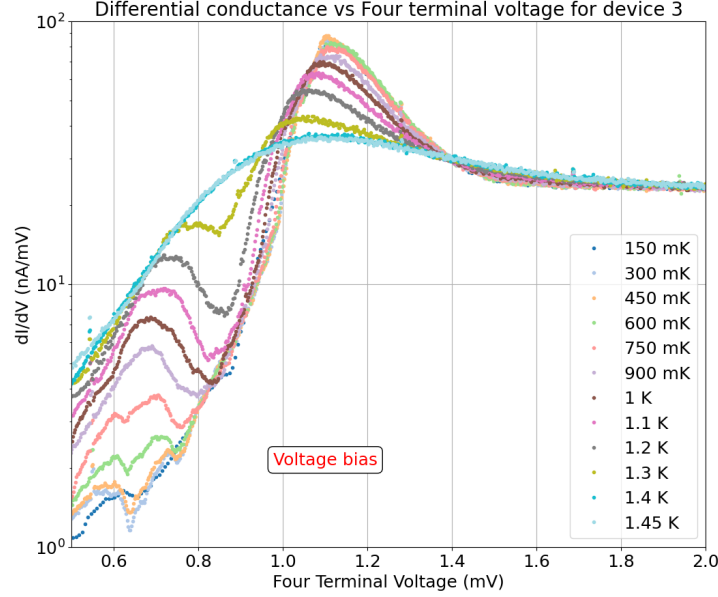
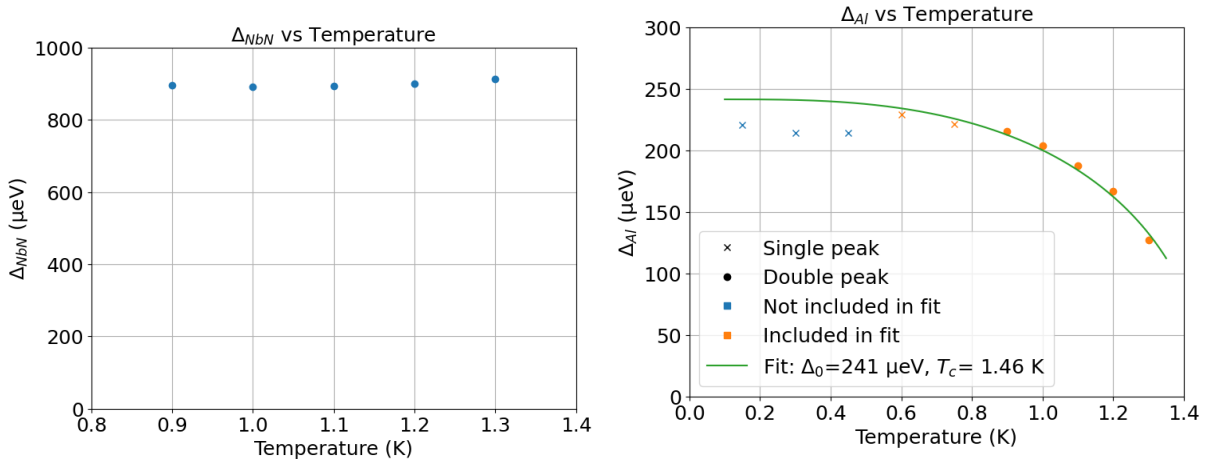


Figure 4.7: Temperature dependence of the differential conductance for device 3 from R4C2.



(a) Temperature dependence of Δ_{NbN} extracted from the peaks in fig. 4.7.

(b) Temperature dependence of Δ_{Al} extracted from the peaks in fig. 4.7. eq. (4.1) is fitted to the data. As can be seen the fit is not very good.

Figure 4.8: Temperature dependence of the two energy gaps present in the asymmetrical JJ. Since NbN has a higher T_c than Al Δ_{NbN} stays constant while Δ_{Al} changes with varying T .

Besides extracting the temperature dependence and values of Δ_{Al} and Δ_{NbN} , the measurements at higher temperature can be used to extract the subgap resistance R_{subgap} , and subsequently the Dynes parameter γ [7]. The peaks in the DOS near the energy gap Δ is broader than expected. This is explained by γ , with a higher value of γ resulting in a wider broadening. A higher values of γ also means a less clean sample with more impurities [8].

When fitting a linear function to the measurements taken at $T = 1.3 \text{ K}$ near where both $I = 0$ and $V = 0$, R_{subgap} can be extracted as the reciprocal of the slope. This is done in fig. 4.9 for device 3 from R2C4. The plot also indicates the cutoff used to distinguish the data

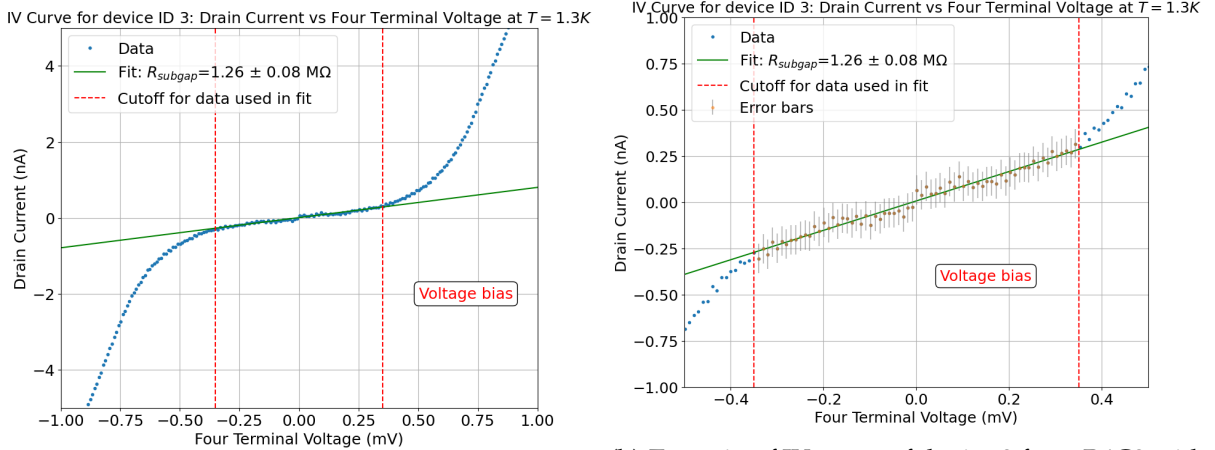
included in the fit. To obtain an uncertainty for R_{subgap} from the fit, the y -values, current, must be assigned appropriate uncertainties. Unlike the voltage, these uncertainties could not be calculated directly. However, since the x -values, voltage, carry an uncertainty, it can be propagated to the y -values using standard error propagation. Because this propagation depends on the fitted value of R_{subgap} , the procedure is iterative and is repeated until the uncertainty converges. The final uncertainties used in the fit are shown in fig. 4.9b. The same procedure was used for device 9.

To calculate γ , the two values of R_n for both devices are combined with a weighted average and then γ is calculated using [7]

$$\gamma = \frac{R_N}{R_{\text{subgap}}}. \quad (4.2)$$

The results are shown in table 4.4. γ is very similar for the two devices, but it differs by a factor 10 from what is obtained in [7].

Importantly, by looking at fig. 4.7 one notices that the two peaks of dI/dV have not completely collapsed for the measurement at $T = 1.3$ K. This means that Aluminum is still superconducting at this point. It first reaches its critical temperature at $T = 1.4$ K. So the measurements used for finding γ should have been collected at $T = 1.4$ K instead.



(a) IV curve of device 3 from chip R4C2 showing both linear and non-linear part.

(b) Zoom in of IV curve of device 3 from R4C2 with error bars showing the final uncertainties used for the fit.

Figure 4.9: IV curve of device 3 from chip R4C2 taken at $T = 1.3$ K used for extracting R_{subgap} .

Table 4.4: Summary of extracted parameters used for calculating the Dynes parameter γ .

ID	R_n [k Ω]	R_{subgap} [M Ω]	γ
3	44.753 ± 0.003	1.26 ± 0.08	35 ± 2
9	32.784 ± 0.003	0.96 ± 0.06	34 ± 2

5 Discussion

Several of the devices were damaged, either during transport or by ESD, to a degree where it was not possible to perform measurements on them. The amplifier and I/V converter started overloading immediately, and no measurements were conducted.

There were also some devices that were left out from the data analysis despite measurements being performed on them. They were not included since their IV curves behaved in a

way that diverged from known theory. There were mainly two types of this behavior.

The first type is shown in fig. 5.1. In the IV curve there are two gaps, but if the junction behaved as a typical JJ, there should only be one. After reaching the first critical current, a new supercurrent runs until reaching a second critical current, after which the junction enters a finite-voltage state. An explanation for this behavior could be that there is another component in series with the JJ which also acts as a superconductor. Once the first critical current is reached one of the superconductors enters a finite-voltage state explaining the non-zero voltage drop. But there is still a superconducting component in series in the circuit since the voltage drop doesn't increase further until the second critical current is reached.

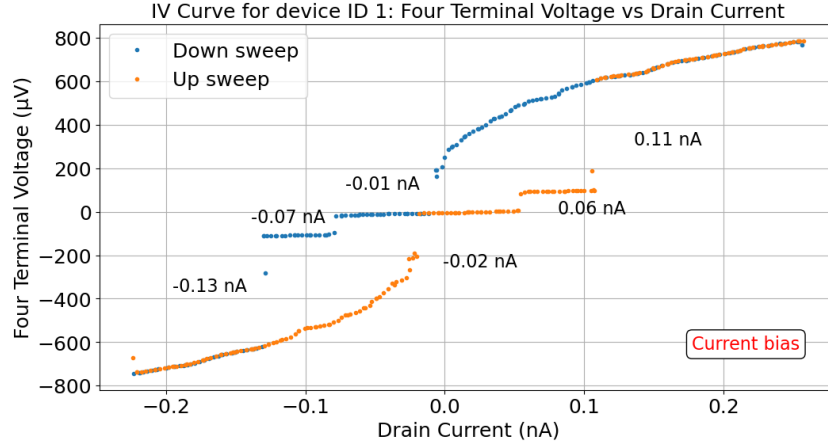
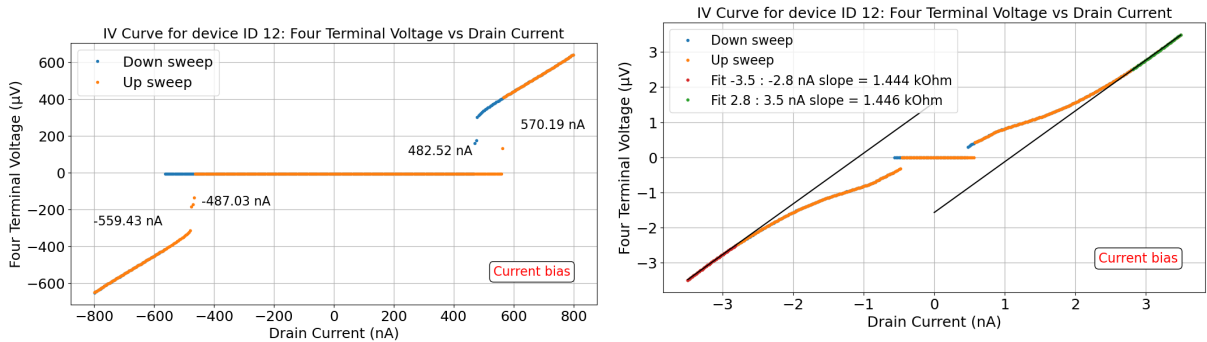


Figure 5.1: IV curve of device 1 from chip R4C2. The JJ is asymmetrical. Here two gaps are present instead of just one as would be expected.

The second type of diverging behavior is shown in fig. 5.2a where the JJ has a very high I_c . This indicates that the JJ has been damaged. It acts more like a SNS-JJ rather than a SIS-JJ. The insulator is ineffective and has presumably been damaged by ESD either during bonding, loading or transportation.



(a) IV curve of device 12 from chip R4C4. Here I_c is very high.

(b) IV curve of device 12 from chip R4C4 used to extract R_N . For this JJ R_N is very small indicating that it has been damaged

Figure 5.2: Odd behavior with very high I_c and low R_n . The JJ measured here is asymmetrical.

For the devices where it was still possible to conduct measurements, device 9 from R2C4 and device 1, 7, and 9 from R4C2 behaved as the example in fig. 5.1 with two critical currents. Device 12 from R4C4 and device 6 and 8 from R4C2 behaved like the example in fig. 5.2 with very high I_c and low R_n . There were some other devices which were possible to measure and did not behave like either of the examples in fig. 5.1 and fig. 5.2a. Their IV curves are shown in Appendix C. No plausible explanations for their behavior was found.

Out of the 31 bonded junctions, 8 showed ideal SIS behavior, 13 were damaged, and 10

did not act according to theory.

6 Conclusion

The purpose of this project was to investigate and evaluate the transport properties of asymmetrical Al/AlOx/NbN JJs at 150 mK. For comparison symmetrical Al/AlOx/Al JJs were also studied. Additional measurements at higher temperatures were also conducted.

The measurements were performed in a Kiutra cryostat and consisted of measuring IV curves through either current biasing or voltage biasing. Current biasing measurements were performed both at low voltages near the energy gap of the JJ Δ/e , and at higher voltages several times larger than Δ/e . The measurements at large voltages were used to extract the normal-state resistance R_n of the JJ. The measurements at low voltages were used to extract Δ/e and I_c . The voltage bias measurements were also used to extract I_c .

For the symmetrical JJs, Δ/e were close in value for different devices. This was not the case for the asymmetrical JJs, with the greatest variation being $\approx 100 \mu V$.

Both up-sweeps with increasing source voltage and down-sweeps with decreasing source voltage were conducted. The values of Δ/e and I_c were similar for up- and down-sweeps for the same device.

The extracted values of I_c from the voltage bias measurements differed from the current bias measurements with all devices, except one, having a greater I_c from voltage bias measurements.

For every device the Ambegaokar-Baratoff relation $I_c R_n \approx 2\Delta/e$ was calculated. For all devices, the relation deviated by one order of magnitude.

The weighted averages of R_n were compared to the room temperature resistance R_{room} , as well as the varying lead width and gap size. All but two values of R_n behaved as expected with a larger lead width and smaller gap size resulting in a smaller R_n .

From the measurements performed at different temperatures the energy gaps of Al and NbN were calculated as $\Delta_{Al} = (241 \pm 2) \mu eV$ and $\Delta_{NbN} = (899 \pm 3) \mu eV$. The critical temperature of Al was found to be $T_c = (1.46 \pm 0.01) K$. Δ_{NbN} stayed constant with varying temperature while Δ_{Al} became smaller with increasing temperature. This is due to NbN having a higher critical temperature than Al.

For device 3 and 9 the Dynes parameter γ was extracted based on measurements performed at $T = 1.3 K$. For device 3 it was $\gamma = 35 \pm 2$ while for device 9 $\gamma = 34 \pm 2$. When looking at the measurements used to obtain the energy gaps of Al and NbN, data shows that the JJ is still a superconductor at $T = 1.3 K$. It is first at $T = 1.4 K$ that it enters a finite-voltage state. Therefore the measurements used for calculating γ should have been performed at $T = 1.4 K$, rather than at $T = 1.3 K$.

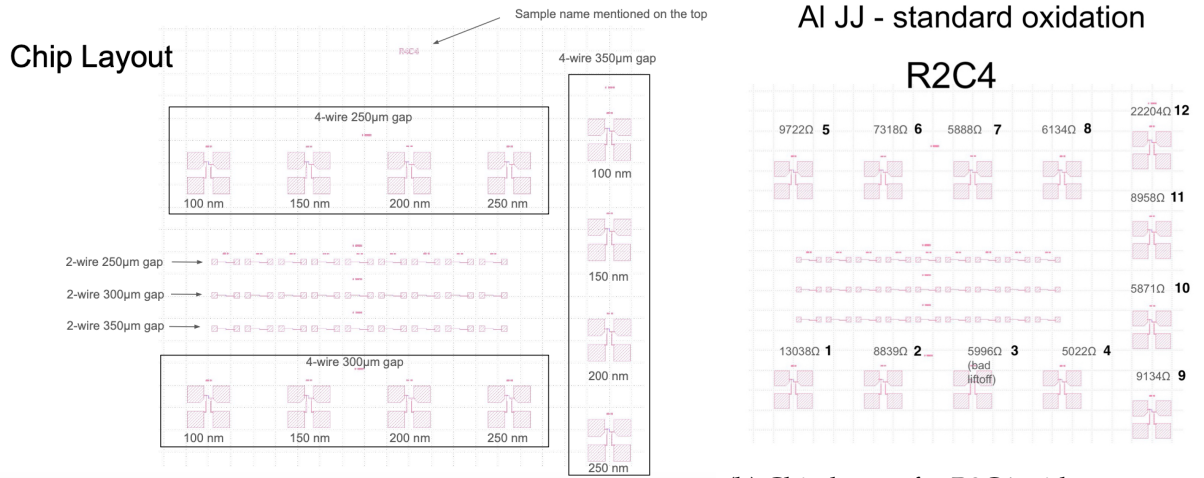
Some of the measured devices behaved in a way that diverged from known theory. There were primarily two types of this behavior. The first was likely caused by another component in series with the JJ which also acted as a superconductor. The second was due to damage to the device. These behaved like SNS-JJ and the insulator in the JJ was ineffective. This is most likely due to damage from ESD either during bonding, loading, or transportation.

For several of the devices it was not possible to conduct measurements, since they had been damaged and proper contact to the device could not be established. Once again most likely due to damage from ESD either during bonding, loading or transportation.

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Appendix A



(a) General chip layout showing the variation in gap size and lead 1 width.

(b) Chip layout for R2C4 with room temperature resistance measures by the fabrication team as well as device IDs.

Figure A.1: General chip layout (a) and layout for R2C4 (b).

Sample 1 & 2 - Asym JJ - standard oxidation



Figure A.2: Chip layout for R4C4 (left) and R4C2 (right) with room temperature resistance measures by the fabrication team as well as device IDs.

Devices fabricated

A total of 6 samples were fabricated. 2 Asym JJ with standard oxidation, 2 Asym JJ with higher oxidation condition and 2 Al JJ with standard oxidation

Sample No	Lead 1 (Al) Thickness	Oxidation parameters	Lead 2 (NbN) Thickness	Lead 2 (Al) Thickness	Comments
1 - R4C2	30 nm	5 Torr 15 mins	40 nm	-	Standard Oxidation
2 - R4C4	30 nm	5 Torr 15 mins	40 nm	-	Standard Oxidation
3 - R3C2	30 nm	20 Torr 20 mins	40 nm	-	Higher oxidation
4 - R3C4	30 nm	20 Torr 20 mins	40 nm	-	Higher oxidation
5 - R2C3	30 nm	5 Torr 15 mins	-	70 nm	Conventional Al JJ
5 - R2C4	30 nm	5 Torr 15 mins	-	70 nm	Conventional Al JJ

Figure A.3: Fabrication details showing the thickness of superconductor and oxidation parameters for each chip.

Appendix B

B.1 Uncertainty on voltage measurements

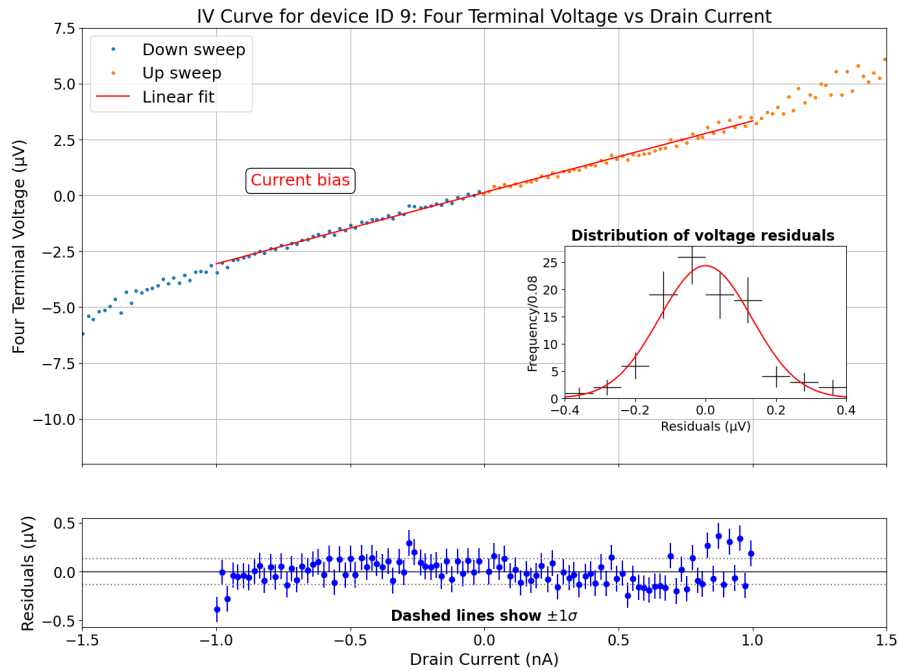


Figure B.1: Plot of the data used to extract an uncertainty on the voltage measurements. The measured voltages between -1 nA and 1 nA are fitted with a linear function. The standard deviation of the residuals is then taken as the error on the measured voltages. Looking at the distribution of the residuals we see that they roughly follow a gaussian so the assumption that they correspond to the error is reasonable.

B.2 Additional IV curves used to extract parameters

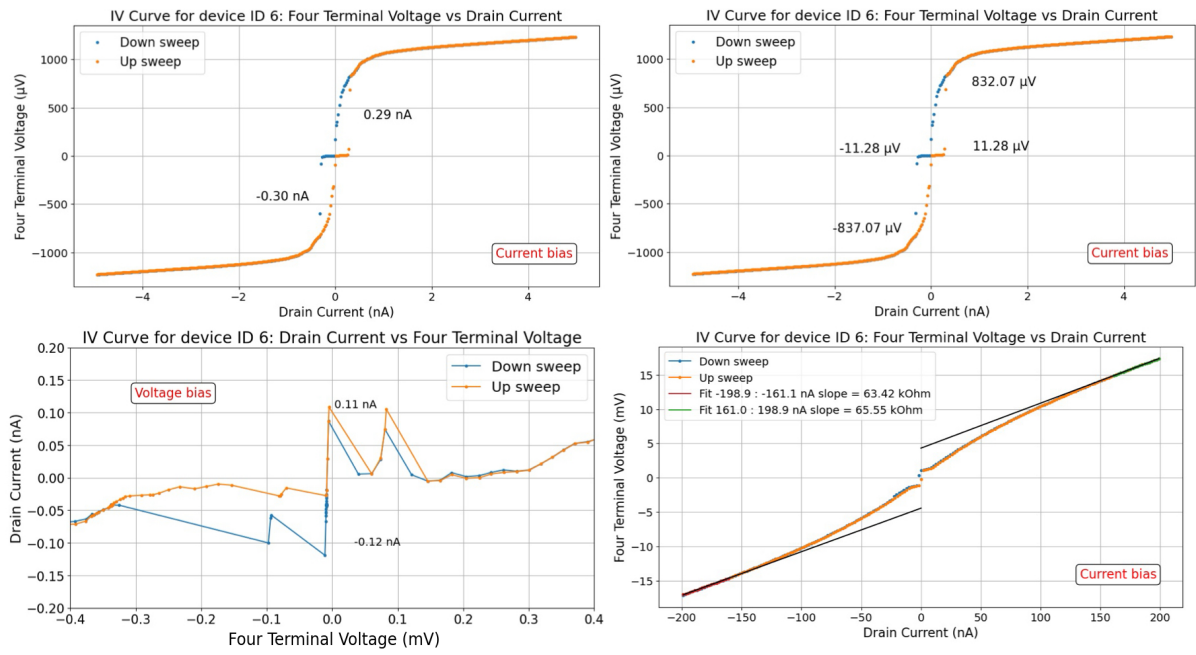


Figure B.2: IV-curves of device 6 from chip R4C4 used for extracting I_c , Δ/e , and R_n . The JJ is an asymmetrical Al/AIOx/NbN junction.

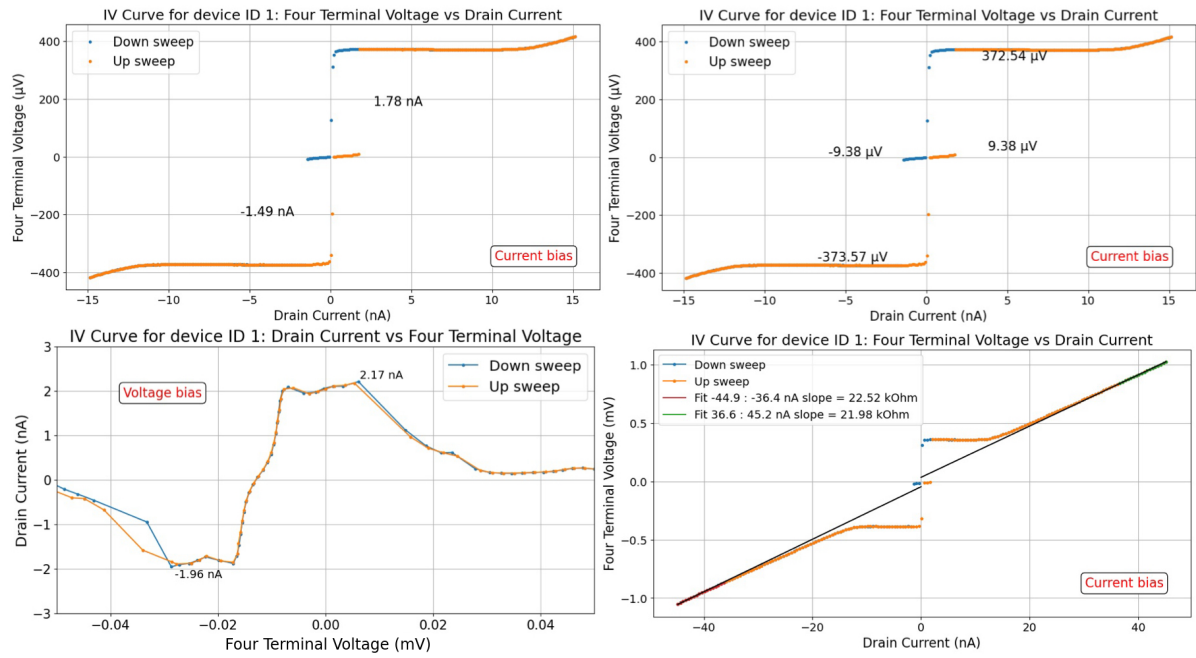


Figure B.3: IV-curves of device 1 from chip R2C4 used for extracting I_c , Δ/e , and R_n . The JJ is a symmetrical Al/AIOx/Al junction.

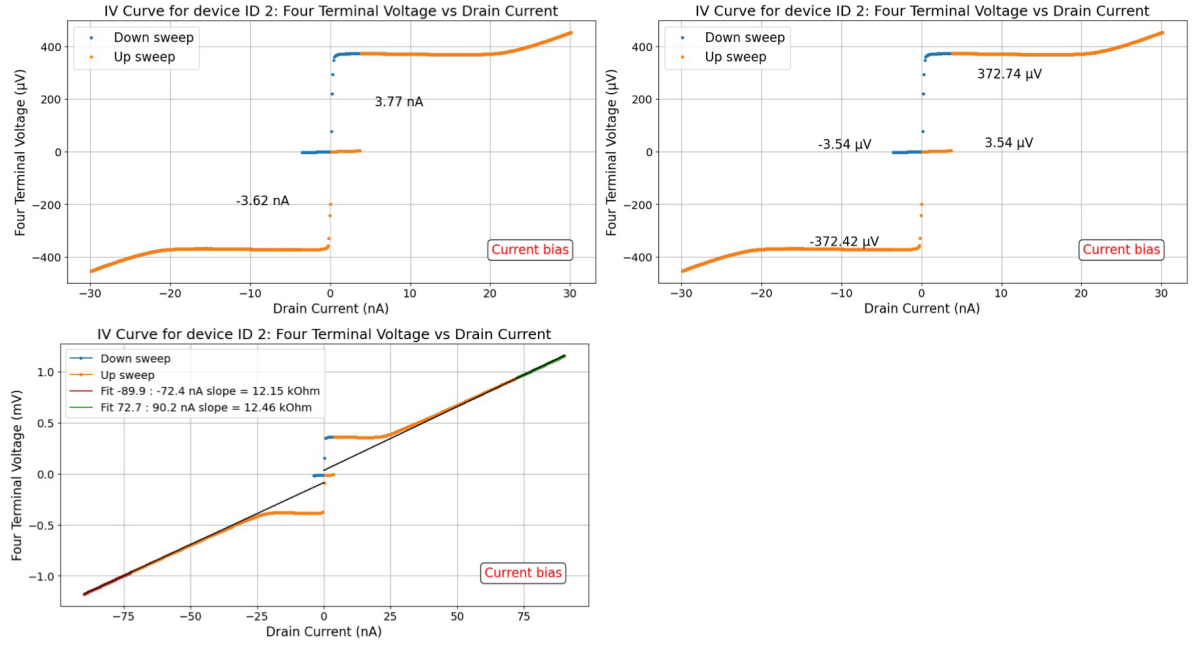


Figure B.4: IV-curves of device 2 from chip R2C4 used for extracting I_c , Δ/e , and R_n . The JJ is a symmetrical Al/AlO_x/Al junction.

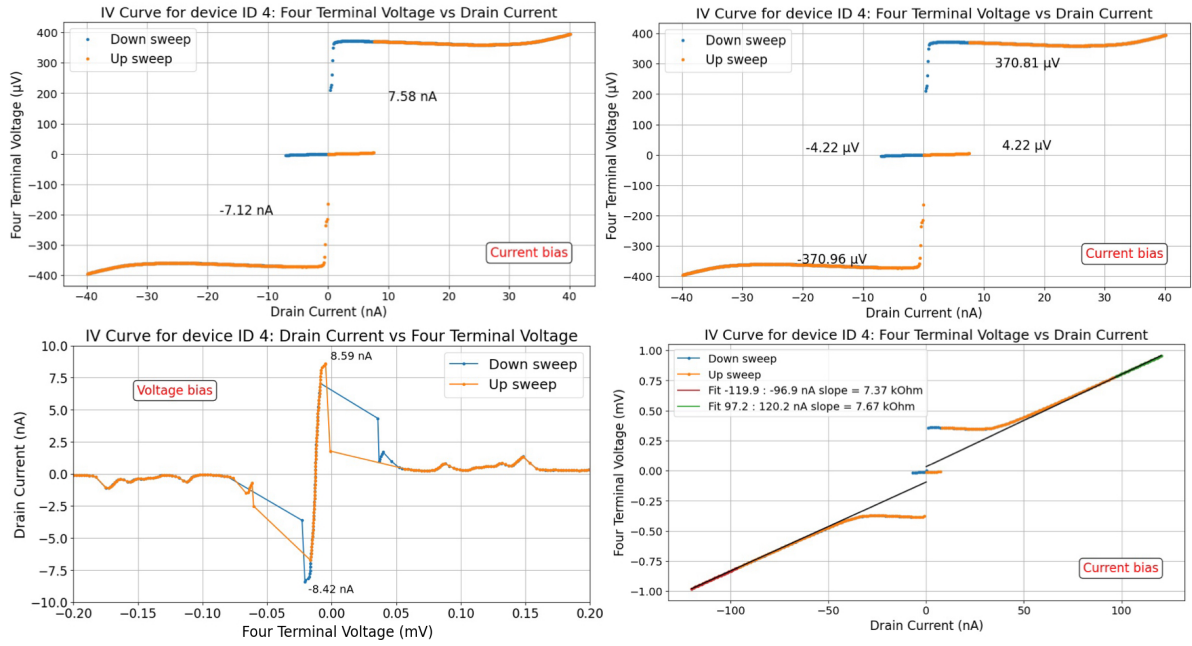


Figure B.5: IV-curves of device 4 from chip R2C4 used for extracting I_c , Δ/e , and R_n . The JJ is a symmetrical Al/AlO_x/Al junction.

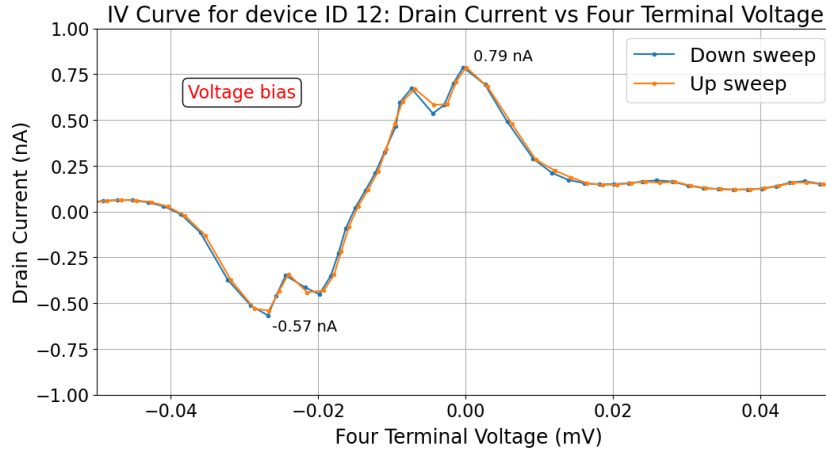


Figure B.6: IV-curves of device 12 from chip R2C4 used for extracting I_c . The JJ is a symmetrical Al/AIOx/Al junction.

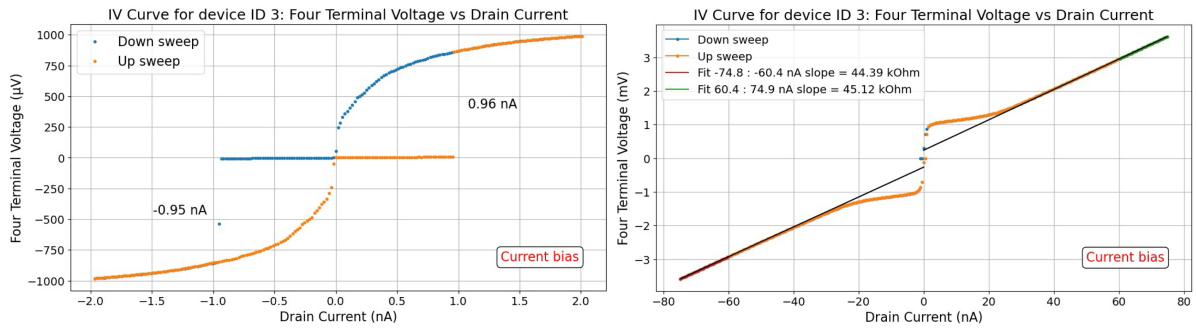


Figure B.7: IV-curves of device 3 from chip R4C2 used for extracting I_c and R_n . The JJ is an asymmetrical Al/AIOx/NbN junction.

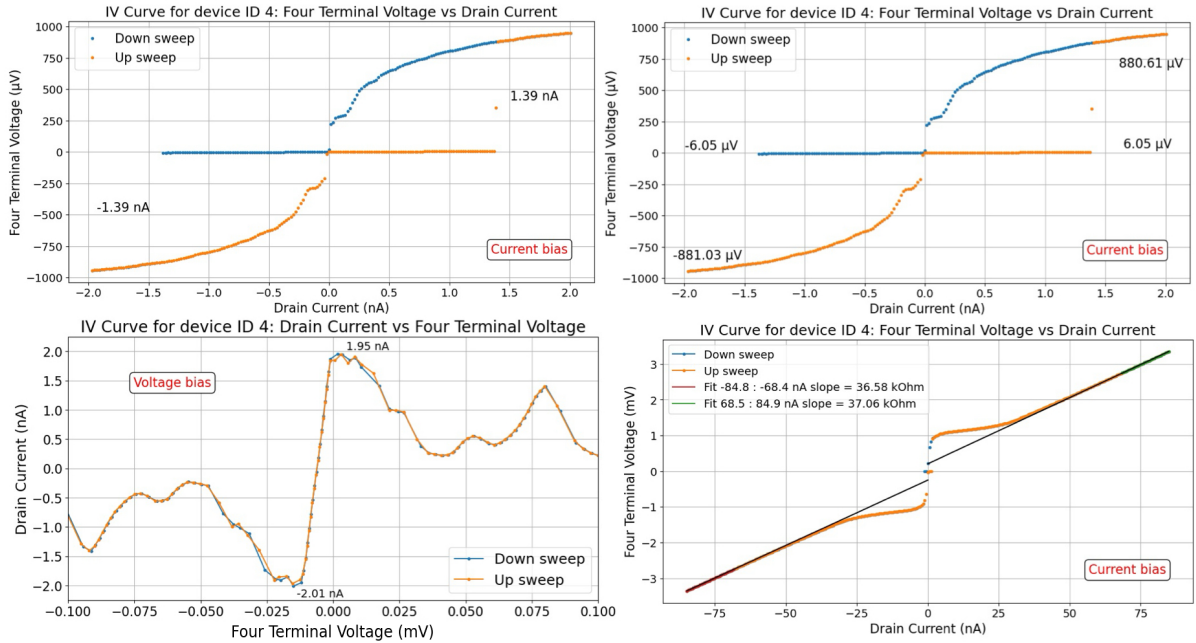


Figure B.8: IV-curves of device 4 from chip R4C2 used for extracting I_c , Δ/e , and R_n . The JJ is an asymmetrical Al/AIOx/NbN junction.

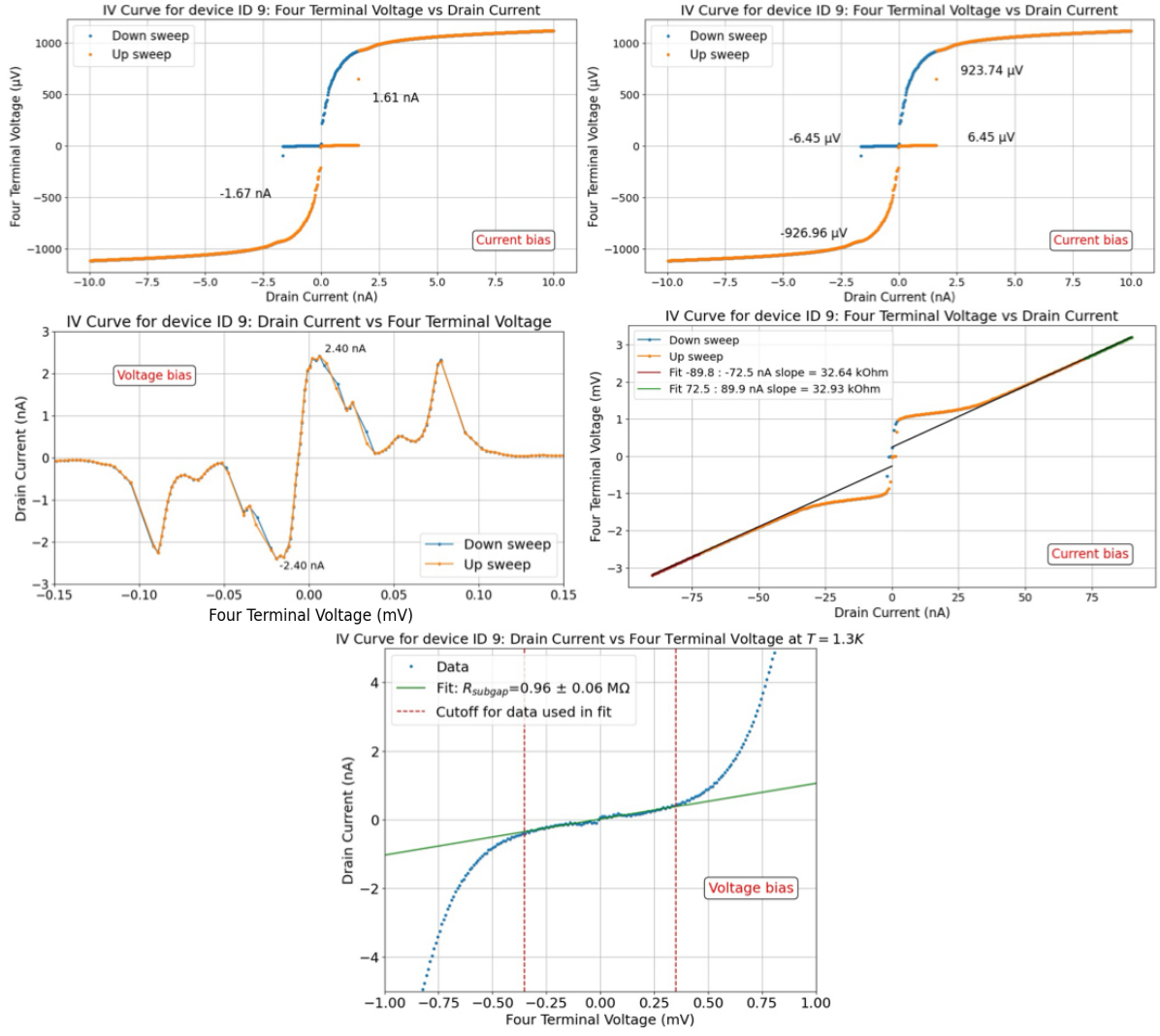


Figure B.9: IV-curves of device 9 from chip R4C2 used for extracting I_c , Δ/e , R_n , and R_{subgap} . The JJ is an asymmetrical Al/AIOx/NbN junction.

Appendix C

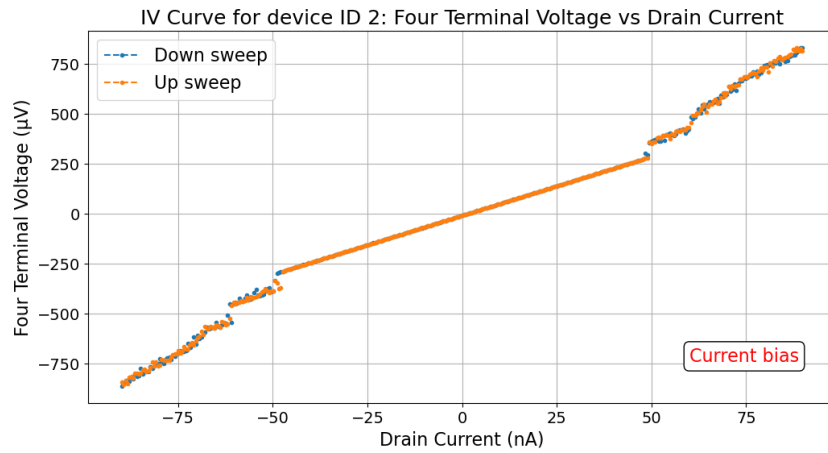


Figure C.1: IV-curve of device 2 from chip R4C4. This measurement was done with a two terminal setup, since two of the four terminals were damaged. It can be calibrated to behave in accordance with theory, which allows for the extraction of Δ/e and I_c . However, the time for doing this was not found during the time frame of the project.

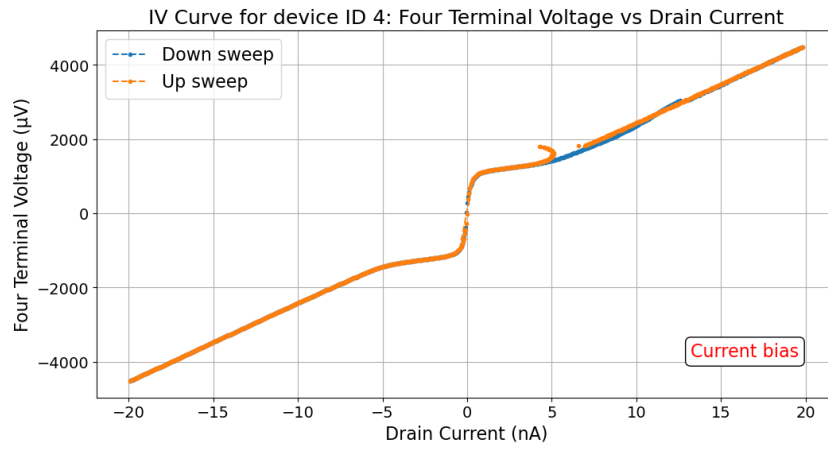


Figure C.2: IV-curve of device 4 from chip R4C4. The characteristics of it differs from theory. A plausible explanation for this behavior was not found.

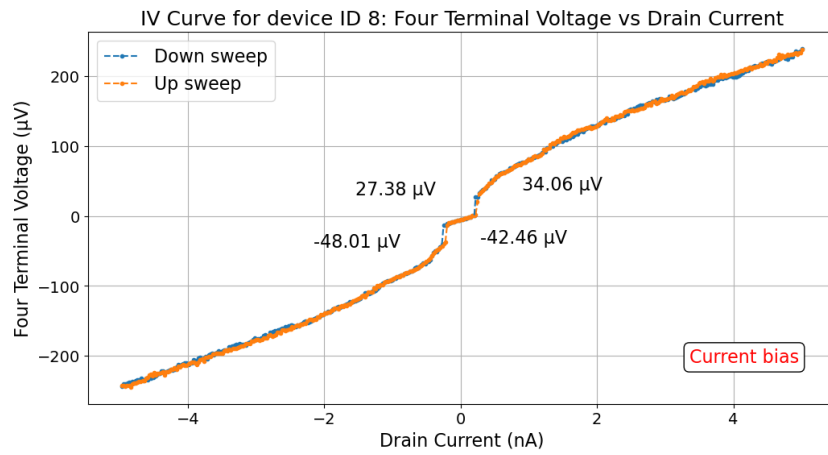


Figure C.3: IV-curve of device 8 from chip R4C4. The characteristics of it differs from theory. A plausible explanation for this behavior was not found.